

A Borel–Lambert Continuation of Ramanujan’s Formula for Odd Zeta Values

Exact lateral Borel–principal-value equivalence and the Lambert-series Stokes jump

Focused working manuscript, 5 October 2026

Abstract

Ramanujan’s self-dual formula expresses $\zeta(4m+3)$ as a finite convolution of even zeta values plus an exponentially convergent Lambert series. Replacing the integer index by a continuous parameter produces a nonterminating factorially divergent convolution. We equip that formal series with an even-factorial Borel transform, continue the transform explicitly to a slit plane, and define its upper and lower lateral Borel–Laplace sums. A beta-integral representation and an explicit kernel K_j identify each lateral sum with a lateral version of an index-independent principal-value integral. Their arithmetic mean is therefore exactly the meromorphic principal-value family. A first-quadrant contour evaluates this family and yields a Borel–Lambert identity, recovers Ramanujan’s terminating class, gives derivative formulas at the complementary odd integers, and shows that the Lambert series is precisely the Stokes jump between the two lateral sums. The underlying continuous Lambert transformation is classical, and related resurgent Lambert-series structures are already known; the contribution claimed here is limited to this even-factorial Borel normalization, its explicit continuation, and the direct Borel–principal-value equivalence in that normalization. The older Barnes finite-part construction is kept separate and remains incompletely specified.

Scope and status. This focused version removes the separate floor-smoothing and $\zeta(5)$ arithmetic programs and retains only the Borel–Lambert continuation. The principal-value identity and the Borel–principal-value identification are proved for the explicitly defined arithmetic mean of the two lateral ray sums. No equivalence with other notions of median resummation is asserted. The Barnes finite-part proposal is not used in the proof. The argument is self-contained at the manuscript level but has not undergone external peer review.

Contents

1	Classical starting point and notation	1
2	Why the moving floor is not the analytic continuation	2
3	The principal-value family	3
4	Explicit continuation of the Borel transform	3
5	Exact Borel–principal-value equivalence	4

6	Contour evaluation and the Borel–Lambert identity	5
7	Ramanujan class, complementary odd values, and residues	6
8	A derived zeta-free cotangent representation	7
9	The Barnes residue kernel and its remaining open problem	8
10	Relation to previous Lambert and resurgent work	8
11	What is established	9
12	Conclusion	9
A	Proof-audit notes for the Borel continuation	9

1 Classical starting point and notation

Define the self-dual Lambert series

$$\mathcal{L}(s) = \sum_{k=1}^{\infty} \frac{k^{-s}}{e^{2\pi k} - 1}. \quad (1.1)$$

The exponential denominator makes (1.1) locally uniformly convergent for all $s \in \mathbb{C}$, so $\mathcal{L}$ is entire. Ramanujan’s self-dual identity is

$$\zeta(4m+3) = -\frac{1}{\pi} \sum_{j=0}^{2m+2} (-1)^j \zeta(2j) \zeta(4m+4-2j) - 2\mathcal{L}(4m+3), \quad m \geq 0. \quad (1.2)$$

For example,

$$\zeta(3) = \frac{7\pi^3}{180} - 2\mathcal{L}(3), \quad \zeta(7) = \frac{19\pi^7}{56700} - 2\mathcal{L}(7).$$

Classical context is reviewed by Berndt and Straub, while generalized Lambert transformations are developed in broader parameter families by Dixit–Maji and Dixit–Gupta–Kumar–Maji [3, 5, 6].

Write

$$s = 4n+3, \quad q = \sin \frac{\pi s}{2}, \quad c = \cos \frac{\pi s}{2}, \quad a_j(s) = -2(-1)^j \zeta(2j) \zeta(s+1-2j). \quad (1.3)$$

At $s = 4m+3$ the coefficient sequence terminates because the shifted zeta factor eventually meets the trivial zeros. For generic s , the functional equation gives

$$a_j(s) = -4c(2\pi)^{s-2j} \Gamma(2j-s) \zeta(2j) \zeta(2j-s), \quad (1.4)$$

which displays factorial growth.

Throughout, PV means symmetric Cauchy principal values at each positive integer pole followed by the limit at infinity. Upper and lower Borel lateral sums are defined by rays of arguments $+\theta$ and $-\theta$ and the limit $\theta \downarrow 0$. Their arithmetic mean is denoted $\mathcal{I}_B$. This precise convention is the only “median” used below.

2 Why the moving floor is not the analytic continuation

A literal real-parameter extension of the terminating convolution is

$$A_J(s) = \sum_{j=0}^J a_j(s), \quad (2.1)$$

$$R_{\text{floor}}(n) = \frac{\cos(2\pi n)}{2\pi} A_{\lfloor 2n+2 \rfloor} (4n+3) - 2\mathcal{L}(4n+3). \quad (2.2)$$

The cutoff changes whenever n crosses a half-integer. The resulting jumps belong to the chosen floor rule, not to an analytically selected continuation. Earlier Fourier experiments found a stabilized period-one waveform with slowly decaying harmonics, behavior naturally explained by jump discontinuities. Agreement with the integer data alone cannot choose a unique analytic interpolation.

The smooth Borel construction begins instead with the formal even transform

$$\mathcal{B}_2(t; s) = \sum_{j \geq 0} \frac{a_j(s)}{(2j)!} t^{2j}. \quad (2.3)$$

The key question is whether its specified lateral Laplace sum can be identified with an independently defined analytic family. That is the equivalence problem addressed next.

3 The principal-value family

For an integer $J \geq 0$, set

$$K_J(x) = \sum_{m \geq 1} \frac{1}{m^{2J}(m^2 - x^2)}, \quad (3.1)$$

$$P_J(s) = \text{PV} \int_0^\infty \frac{x^{2J+1-s} K_J(x)}{e^{2\pi x} - 1} dx, \quad (3.2)$$

$$\tilde{I}_J(s) = A_J(s) - 4cP_J(s). \quad (3.3)$$

Near zero the integrand in (3.2) is $O(x^{2J-\Re s})$, so P_J is holomorphic for $\Re s < 2J+1$. At infinity the exponential denominator makes the symmetric pole-cell contributions locally uniformly summable.

Lemma 3.1 (Independence of the subtraction index). *On the overlap $\Re s < 2J+1$, the meromorphic functions satisfy $\tilde{I}_{J+1}(s) = \tilde{I}_J(s)$.*

Proof. The kernel recursion

$$x^2 K_{J+1}(x) = K_J(x) - \zeta(2J+2)$$

gives

$$P_{J+1} - P_J = -\zeta(2J+2) \frac{\Gamma(2J+2-s)\zeta(2J+2-s)}{(2\pi)^{2J+2-s}}.$$

By (1.4), $a_{J+1}(s) = 4c(P_{J+1} - P_J)$. Since $A_{J+1} - A_J = a_{J+1}$, the difference in (3.3) vanishes. Exceptional parameters are taken meromorphically. $\square$

The compatible half-planes exhaust $\mathbb{C}$ and define one meromorphic family, denoted $\tilde{I}(s)$.

4 Explicit continuation of the Borel transform

The missing step in the earlier principal-value manuscript was an explicit continuation of (2.3) with enough control to justify the lateral Laplace limit. The following formula supplies it.

Let

$$\Sigma_J = \{s \in \mathbb{C} : -1 < \Re s < 2J + 1\}, \quad \Omega = \mathbb{C} \setminus ((-\infty, -2\pi] \cup [2\pi, \infty)).$$

For $t \in \Omega$ and $u \in (0, 1)$ put

$$X_k(t, u) = \frac{tu}{2\pi k}.$$

Theorem 4.1 (Explicit Borel continuation). *For $s \in \Sigma_J$ away from coefficient poles, the germ (2.3) extends from $|t| < 2\pi$ to Ω by*

$$\begin{aligned} \widehat{\mathcal{B}}_2(t; s) = & \sum_{j=0}^J \frac{a_j(s)t^{2j}}{(2j)!} \\ & - \frac{4c(2\pi)^s}{\Gamma(s+1)} \int_0^1 u^{-s-1}(1-u)^s \sum_{k \geq 1} k^s X_k(t, u)^{2J+2} K_J(X_k(t, u)) du. \end{aligned} \quad (4.1)$$

On every closed subsector of the right half-plane avoiding the positive real cut, this continuation has at most polynomial growth in t .

Proof. For $j \geq J + 1$, the strip Σ_J gives $\Re(2j - s) > 1$. The beta identity

$$\frac{\Gamma(2j - s)}{(2j)!} = \frac{1}{\Gamma(s+1)} \int_0^1 u^{2j-s-1}(1-u)^s du \quad (4.2)$$

may therefore be inserted into (1.4). Expanding $\zeta(2j - s) = \sum_{k \geq 1} k^{s-2j}$ and using

$$\sum_{j \geq J+1} \zeta(2j) x^{2j} = x^{2J+2} K_J(x), \quad |x| < 1, \quad (4.3)$$

yields (4.1) for $|t| < 2\pi$.

The right side of (4.1) itself provides the continuation. For compact subsets of $\Sigma_J \times \Omega$, the k -sum is locally uniformly convergent: for large k , $X_k \rightarrow 0$ and the summand is $O(k^{\Re s - 2J - 2})$. At $u = 0$ the full integrand is $O(u^{2J+1-\Re s})$, while at $u = 1$ it is $O((1-u)^{\Re s})$. These are integrable precisely on the stated strip. The poles of K_J are reached only when t lies on one of the two removed real rays. Analyticity follows by locally uniform convergence. If two admissible indices J and J' are used, their formulas agree in $|t| < 2\pi$ because both equal the same Borel germ; uniqueness of analytic continuation on the connected slit plane therefore makes the continuation independent of J .

For the growth statement, the partial-fraction formula for K_0 and the recursion in Lemma 3.1 give polynomial bounds for K_J on closed sectors separated from the real pole rays. Substitution into (4.1) gives polynomial growth, uniformly on compact s -subsets of Σ_J . $\square$

5 Exact Borel–principal-value equivalence

For $0 < \theta < \pi/2$, let $\Gamma_{\pm\theta}$ be the rays $re^{\pm i\theta}$, $r \geq 0$, and define

$$\mathcal{S}_{\pm, \theta}(s) = \int_{\Gamma_{\pm\theta}} e^{-t} \widehat{\mathcal{B}}_2(t; s) dt.$$

Polynomial growth from Theorem 4.1 and $\Re t > 0$ make these integrals absolutely convergent.

Theorem 5.1 (Lateral sums and their median). *For $s \in \Sigma_J$ away from coefficient poles,*

$$\mathcal{S}_{\pm, \theta}(s) = A_J(s) - 4c \int_0^{\infty e^{\pm i\theta}} \frac{x^{2J+1-s} K_J(x)}{e^{2\pi x} - 1} dx. \quad (5.1)$$

The limits $\mathcal{S}_+(s) = \lim_{\theta \downarrow 0} \mathcal{S}_{+, \theta}(s)$ and $\mathcal{S}_-(s) = \lim_{\theta \downarrow 0} \mathcal{S}_{-, \theta}(s)$ exist and satisfy

$$\boxed{\mathcal{S}_+(s) = \tilde{I}(s) - 2\pi i c \mathcal{L}(s), \quad \mathcal{S}_-(s) = \tilde{I}(s) + 2\pi i c \mathcal{L}(s).} \quad (5.2)$$

Consequently the arithmetic mean of the two lateral Borel sums is exactly

$$\boxed{\mathcal{I}_B(s) = \frac{\mathcal{S}_+(s) + \mathcal{S}_-(s)}{2} = \tilde{I}(s).} \quad (5.3)$$

The identity extends meromorphically to all s .

Proof. The polynomial part of (4.1) contributes $A_J(s)$ because $\int_{\Gamma_{\pm\theta}} e^{-t} t^{2j} dt = (2j)!$. In the tail, for each k and u set $t = 2\pi kx/u$. The ray for x has the same argument as the ray for t . Absolute convergence permits the interchanges on compact substrips, and the remaining u -integral is

$$\int_0^1 u^{-s-2} (1-u)^s e^{-2\pi kx/u} du = e^{-2\pi kx} \frac{\Gamma(s+1)}{(2\pi kx)^{s+1}}, \quad (5.4)$$

obtained by $y = (1-u)/u$. All powers use the branch inherited from the ray. The factors cancel exactly, and summing $e^{-2\pi kx}$ over k gives (5.1).

The x -integrand in (5.1) has simple poles at the positive integers. Since

$$\operatorname{Res}_{x=m} K_J(x) = -\frac{1}{2m^{2J+1}},$$

its residue at m is

$$-\frac{1}{2} \frac{m^{-s}}{e^{2\pi m} - 1}. \quad (5.5)$$

The residue series is absolutely and locally uniformly convergent in s . Splitting the ray integral at a large finite M , taking the upper or lower limit around the finitely many poles below M , and then letting $M \rightarrow \infty$ gives

$$\begin{aligned} \int_0^{\infty e^{+i\theta}} \frac{x^{2J+1-s} K_J(x)}{e^{2\pi x} - 1} dx &= P_J(s) + \frac{i\pi}{2} \mathcal{L}(s), \\ \int_0^{\infty e^{-i\theta}} \frac{x^{2J+1-s} K_J(x)}{e^{2\pi x} - 1} dx &= P_J(s) - \frac{i\pi}{2} \mathcal{L}(s). \end{aligned}$$

Substitution into (5.1) proves (5.2) and (5.3). Lemma 3.1 then supplies the meromorphic continuation. $\square$

Corollary 5.2 (Stokes jump). *With the stated lateral convention,*

$$\mathcal{S}_-(s) - \mathcal{S}_+(s) = 4\pi i c \mathcal{L}(s). \quad (5.6)$$

Thus the Lambert series measures the jump between the two lateral sums, while their arithmetic mean removes it.

Remark 5.3. *Theorem 5.1 closes the specific analytic gap that remained in the corrected principal-value manuscript: (4.1) is an explicit analytic continuation of the Borel transform, its sectorial growth is sufficient for the ray Laplace integrals, and (5.1) identifies the lateral limits with the principal-value prescription. It does not identify the result with a differently defined nonlinear median or with the older Barnes finite part.*

6 Contour evaluation and the Borel–Lambert identity

The equivalence theorem reduces the Borel problem to evaluating $\tilde{I}$. For $\Re s < -1$ define

$$M(s) = \int_0^\infty \frac{x^{-s}}{e^{2\pi x} - 1} dx = \frac{\Gamma(1-s)\zeta(1-s)}{(2\pi)^{1-s}}, \quad (6.1)$$

$$B(s) = \text{PV} \int_0^\infty \frac{x^{-s} \cot(\pi x)}{e^{2\pi x} - 1} dx. \quad (6.2)$$

The partial-fraction identity

$$K_0(x) = \frac{1}{2x^2} - \frac{\pi \cot(\pi x)}{2x}$$

and the zeta functional equation give

$$\tilde{I}(s) = 2\pi c B(s), \quad \Re s < -1. \quad (6.3)$$

To evaluate $B(s)$, use the first-quadrant function

$$G_s(z) = \frac{z^{-s}}{(e^{2\pi z} - 1)(e^{-2\pi iz} - 1)}, \quad 0 \leq \arg z \leq \frac{\pi}{2}. \quad (6.4)$$

Integrate around a first-quadrant contour with a small circle at the origin, symmetric clockwise indentations at the poles on both axes, and outer radius $N + 1/2$. The outer arc decays exponentially. Writing $E = e^{-i\pi s/2}$, the real-axis contribution is $(-M + iB)/2$, while the upward imaginary-axis contribution is $iE(-M - iB)/2$. The boundary residues are

$$\text{Res}_{z=m} G_s = \frac{i}{2\pi} \frac{m^{-s}}{e^{2\pi m} - 1}, \quad \text{Res}_{z=im} G_s = \frac{E}{2\pi} \frac{m^{-s}}{e^{2\pi m} - 1}.$$

After summing the exponentially convergent residue series,

$$B(s) = \frac{1-q}{c} [\mathcal{L}(s) - M(s)]. \quad (6.5)$$

Since the functional equation also gives $\zeta(s) = 2qM(s)$, the identity theorem and Lemma 3.1 yield the central formula.

Theorem 6.1 (Borel–Lambert identity). *For the arithmetic mean of lateral Borel sums defined above, as a meromorphic identity,*

$$\boxed{\mathcal{I}_B(s) = -\pi \frac{1-q}{q} \zeta(s) + 2\pi(1-q)\mathcal{L}(s).} \quad (6.6)$$

Equivalently,

$$\boxed{\zeta(s) = -\frac{q}{\pi(1-q)} \mathcal{I}_B(s) + 2q\mathcal{L}(s).} \quad (6.7)$$

At trigonometric exceptional points, the complete expressions are interpreted by meromorphic limits.

7 Ramanujan class, complementary odd values, and residues

At $s = 4m + 3$, the formal series terminates. Then $\mathcal{I}_B = A_J$ for all sufficiently large J , and (6.7) is exactly Ramanujan's identity (1.2).

At the complementary odd class $s_0 = 4m + 1 > 1$, $q(s_0) = 1$ and $1 - q(s) = \pi^2(s - s_0)^2/8 + O((s - s_0)^4)$. Hence

$$\mathcal{I}_B(s_0) = \mathcal{I}'_B(s_0) = 0, \quad (7.1)$$

and comparison of the quadratic coefficients in (6.6) gives

$$\boxed{\zeta(s_0) = 2\mathcal{L}(s_0) - \frac{4}{\pi^3}\mathcal{I}''_B(s_0).} \quad (7.2)$$

Thus the earlier numerical observations $\mathcal{I}_B(5) = 0$ and $\mathcal{I}'_B(5) = 0$ are exact consequences of the identity, not independent conjectures. At $s = 1$, the pole of zeta changes the order and

$$\mathcal{I}_B(s) = -\frac{\pi^3}{8}(s - 1) + O((s - 1)^2). \quad (7.3)$$

At nonnegative even integers $s = 2r$,

$$\operatorname{Res}_{s=2r} \mathcal{I}_B(s) = 2(-1)^{r+1}\zeta(2r). \quad (7.4)$$

At negative even integers the trivial zero cancels the apparent singularity.

In the original n -parameter, define

$$R_B(n) = \frac{\cos(2\pi n)}{2\pi} \mathcal{I}_B(4n + 3) - 2\mathcal{L}(4n + 3). \quad (7.5)$$

Since $q = -\cos(2\pi n)$, Theorem 6.1 gives

$$\boxed{R_B(n) = \zeta(4n + 3) \cos^2(\pi n) - 2\mathcal{L}(4n + 3)Q(n),} \quad (7.6)$$

where

$$Q(n) = \sin^2(\pi n)[1 + 2\cos^2(\pi n)] = \frac{3}{4} - \frac{1}{2}\cos(2\pi n) - \frac{1}{4}\cos(4\pi n). \quad (7.7)$$

This formula began as a numerical correction to the false model $\zeta(4n + 3) \cos^2(\pi n)$; it is now a theorem under the stated Borel prescription.

For $n_0 = m + 1/2$, the factor $\cos^2(\pi n)$ has a double zero. If $F(n) = R_B(n) + 2\mathcal{L}(4n + 3)Q(n)$, then $F(n) = \zeta(4n + 3) \cos^2(\pi n)$ and

$$\zeta(4m + 5) = \frac{F''(n_0)}{2\pi^2} = 2\mathcal{L}(4m + 5) - \frac{1}{4\pi^3} \left. \frac{d^2}{dn^2} \mathcal{I}_B(4n + 3) \right|_{n=n_0}. \quad (7.8)$$

This is a derivative identity, not a recurrence in m .

8 A derived zeta-free cotangent representation

The contour proof itself gives a representation containing no zeta values on the right. Combining (6.3) with (6.7) yields

$$\zeta(s) = 2q \left[\mathcal{L}(s) - \frac{c}{1-q} \text{PV} \int_0^\infty \frac{x^{-s} \cot(\pi x)}{e^{2\pi x} - 1} dx \right]. \quad (8.1)$$

It is initially obtained for $\Re s < -1$ and then continued meromorphically. The quotient is not numerically well-conditioned near every exceptional value, so (8.1) is best viewed as an analytic consequence of the framework rather than the preferred universal evaluator.

For an explicit continuation of the cotangent integral, one may expand near zero

$$\frac{\cot(\pi x)}{e^{2\pi x} - 1} = \sum_{k \geq 0} h_k x^{k-2}, \quad h_k = \frac{(2\pi)^k}{2\pi^2} \sum_{0 \leq 2j \leq k} \frac{(-1)^j B_{2j} B_{k-2j}}{(2j)!(k-2j)!}. \quad (8.2)$$

For $0 < \alpha < 1$,

$$B(s) = \sum_{k \geq 0} \frac{h_k \alpha^{k-1-s}}{k-1-s} + \text{PV} \int_\alpha^\infty \frac{x^{-s} \cot(\pi x)}{e^{2\pi x} - 1} dx. \quad (8.3)$$

The series has geometric tails on compact sets after choosing $\alpha < \rho < 1$ inside the radius of convergence. Apparent poles from separate pieces are always interpreted in the combined meromorphic expression.

9 The Barnes residue kernel and its remaining open problem

The natural residue kernel

$$\mathcal{K}(z; s) = -\pi \csc(\pi z/2) \zeta(z) \zeta(s+1-z) \quad (9.1)$$

satisfies

$$\text{Res}_{z=2j} \mathcal{K}(z; s) = a_j(s). \quad (9.2)$$

Thus finite contours enclosing $0, 2, \dots, 2N$ reproduce the finite convolution. An earlier proposal wrote schematically

$$J_B(s) \stackrel{\text{formal}}{=} \text{med FP}_{N \rightarrow \infty} \frac{1}{2\pi i} \int_{C_N} \mathcal{K}(z; s) dz. \quad (9.3)$$

This remains insufficiently specified. One must prescribe the contour geometry, subtraction, collisions with the poles at $z = 1$ and $z = s$, the limiting order, and deformation invariance. The excluded residues include

$$\text{Res}_{z=1} \mathcal{K} = -\pi \zeta(s), \quad \text{Res}_{z=s} \mathcal{K} = \pi \csc(\pi s/2) \zeta(s).$$

Accordingly, the Barnes kernel is a useful structural explanation but is no longer needed to prove the Borel equivalence: Theorem 5.1 proves that equivalence directly from the Borel transform. Equality of the still-undefined finite part (9.3) with $\mathcal{I}_B$ remains a separate open problem.

10 Relation to previous Lambert and resurgent work

The literature contains broad families of Ramanujan-type and generalized Lambert transformations [3, 5, 6]. In particular, the Abel–Plana/Ramanujan framework developed by Berndt–Dixit–Gupta–Zaharescu contains a continuous-parameter transformation which specializes to Ramanujan’s odd-zeta formula [11]. Consequently, the principal-value contour identity used in Section 6 should be viewed as a realization, in the present notation, of classical transformation machinery rather than as a new Lambert identity. Before publication the exact printed form of the relevant notebook entry should be checked line-by-line against the primary source.

There is also close resurgent precedent. Dorigoni and Kleinschmidt study Lambert-series transseries whose perturbative coefficients, after use of the functional equation value $\zeta(1 - 2j)$, match the present coefficient family up to the normalization factor associated with the even-factorial Borel transform [12]. Their Stokes discontinuity likewise reduces to a divisor-sum Lambert series which, at the self-dual point, is the same $\mathcal{L}(s)$ up to normalization and sign convention. The Borel/median/Stokes mechanism is therefore not claimed as a new phenomenon.

The distinctive content retained here is narrower: (i) the direct even-factorial Borel normalization of Ramanujan’s convolution; (ii) the explicit K_J -kernel continuation and its J -independence argument; and (iii) the direct proof that the arithmetic mean of the two lateral Borel–Laplace sums equals the meromorphic principal-value family, together with the resulting Lambert-series Stokes jump. These are formulation and equivalence results, not priority claims over the classical Lambert transformation or over general resurgence theory.

The results here do not remove the Lambert contribution from Ramanujan’s formulas and do not imply new irrationality or transcendence results for odd zeta values.

11 What is established

Statement	Status
Explicit continuation of the even-factorial Borel transform	Proved on the slit plane by Theorem 4.1, first on the strip $-1 < \Re s < 2J + 1$ and then continued meromorphically.
Existence of the specified upper and lower lateral Borel–Laplace sums	Proved from the continuation and sectorial polynomial growth.
Arithmetic-mean Borel sum $\mathcal{I}_B = \tilde{\mathcal{I}}$	Proved by the ray change of variables and symmetric pole limits.
Individual lateral jump	Proved: $\mathcal{S}_- - \mathcal{S}_+ = 4\pi i \cos(\pi s/2) \mathcal{L}(s)$.
Borel–Lambert closed form and zeta isolation	Proved meromorphically.
Terminating Ramanujan class	Classical and recovered directly from the Borel identity.
Complementary odd derivative formulas	Consequences of the proved Borel–Lambert identity.
Older Barnes finite part	Not fully defined; equivalence with $\mathcal{I}_B$ remains open.
Broad novelty of the Lambert transformation or Stokes mechanism	Not claimed.
New irrationality or transcendence results	None.

12 Conclusion

The analytic continuation problem can be stated cleanly without the auxiliary smoothing or arithmetic investigations. Ramanujan's terminating convolution extends formally to a factorially divergent even series. The explicit continuation formula in Theorem 4.1 turns its chosen Borel germ into an analytic function on a slit plane with sufficient sectorial growth for lateral Laplace summation. Theorem 5.1 then gives the central identification: the arithmetic mean of the upper and lower lateral Borel sums is exactly the index-independent principal-value family.

The contour evaluation in Section 6 converts that family into the Borel–Lambert identity. In this normalization, the exponentially convergent Lambert series has a precise resurgent role: it is the discontinuity between the two lateral sums. Ramanujan's original $s = 4m + 3$ formula is recovered where the formal convolution terminates, while the complementary odd class follows from the double zero of the Borel sum.

The literature comparison narrows the interpretation. Continuous Ramanujan–Lambert transformations are classical, and closely related Stokes/Lambert structures occur in earlier resurgent work. The point of the present paper is therefore not the discovery of the transformation itself, but the explicit even-factorial Borel normalization, the K_J continuation that glues the subtraction indices, and the direct proof that this Borel prescription equals the principal-value family. The remaining analytic open problem internal to this manuscript is the older Barnes finite-part construction, whose residue kernel is natural but whose finite-part prescription is still not complete.

A Proof-audit notes for the Borel continuation

The proof of Theorem 5.1 uses four points that are worth isolating for later audit.

1. The beta integral (4.2) is used only on $-1 < \Re s < 2J + 1$, where both endpoint exponents are integrable and $\Re(2j - s) > 1$ for every $j \geq J + 1$.
2. The continuation formula (4.1) is first derived in the disk $|t| < 2\pi$, where all power-series manipulations are absolutely convergent. The right side then defines the continuation on the slit plane by locally uniform convergence.
3. For fixed nonzero ray angle, the Laplace integral is absolutely convergent. On such a ray one has $|m^2 - X^2| \geq C_\theta(m^2 + |X|^2)$, so $K_J(X)$ has a uniform algebraic bound. Splitting the k -sum at $k \asymp |t|u$ gives at most polynomial growth in $|t|$, while $e^{-\Re t}$ supplies exponential decay. The endpoint factors are integrable on the strip. Thus Tonelli–Fubini applies on compact substrips. After $t = 2\pi kx/u$, the identity (5.4) and the geometric sum $\sum_{k \geq 1} e^{-2\pi kx} = (e^{2\pi x} - 1)^{-1}$ give (5.1) without taking a singular real-axis limit prematurely.
4. The upper/lower limits are taken after splitting at finitely many poles. The remaining residue tail is uniformly negligible because (5.5) is exponentially summable. Their arithmetic mean is therefore the symmetric principal value.

These statements establish the equivalence for the particular lateral-ray Borel prescription used in this paper. A broader resurgent interpretation would require additional definitions and is not claimed.

References

- [1] W. Balser, *From Divergent Power Series to Analytic Functions: Theory and Application of Multi-summable Power Series*, Lecture Notes in Mathematics 1582, Springer, 1994.

- [2] B. C. Berndt, *Ramanujan's Notebooks, Part II*, Springer-Verlag, New York, 1989.
- [3] B. C. Berndt and A. Straub, "Ramanujan's formula for $\zeta(2n + 1)$," in *Exploring the Riemann Zeta Function*, Springer, 2017, pp. 13–34; <https://arxiv.org/abs/1701.02964>.
- [4] O. Costin, *Asymptotics and Borel Summability*, CRC Press, 2008.
- [5] A. Dixit and B. Maji, "Generalized Lambert series and arithmetic nature of odd zeta values," *Proceedings of the Royal Society of Edinburgh Section A* **150** (2020), 741–769. doi:10.1017/prm.2018.146; <https://arxiv.org/abs/1709.00022>.
- [6] A. Dixit, R. Gupta, R. Kumar, and B. Maji, "Generalized Lambert series, Raabe's integral and a two-parameter generalization of Ramanujan's formula for $\zeta(2m + 1)$," arXiv:1801.09181, 2018; <https://arxiv.org/abs/1801.09181>.
- [7] NIST Digital Library of Mathematical Functions, Chapter 25, especially §25.5, Eq. 25.5.11, and the functional equations in §25.4, <https://dlmf.nist.gov/25.5.E11>, accessed September 2026.
- [8] R. B. Paris and D. Kaminski, *Asymptotics and Mellin–Barnes Integrals*, Encyclopedia of Mathematics and its Applications 85, Cambridge University Press, 2001.
- [9] S. Ramanujan, *The Lost Notebook and Other Unpublished Papers*, Narosa, New Delhi, 1988.
- [10] A. D. Sokal, "An improvement of Watson's theorem on Borel summability," *Journal of Mathematical Physics* **21** (1980), 261–263. doi:10.1063/1.524408.
- [11] B. C. Berndt, A. Dixit, R. Gupta, and A. Zaharescu, "Ramanujan and Koshliakov Meet Abel and Plana," arXiv:2112.09819, 2021; <https://arxiv.org/abs/2112.09819>.
- [12] D. Dorigoni and A. Kleinschmidt, "Resurgent expansion of Lambert series and iterated Eisenstein integrals," *Communications in Number Theory and Physics* **15** (2021), 1–57; arXiv:2001.11035; <https://arxiv.org/abs/2001.11035>.